

Ali Meski

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EDUCATION

Bachelor of Engineering, Systems and Computing Engineering (Co-op)

2024–Present

University of Guelph, Guelph, Ontario

- Dean's Honours List – Winter 2025; Entrance Scholarship of \$2,000 for 93% admission average
- Relevant Courses: Data Structures, Digital System Design, Object-Oriented Programming

SKILLS

Embedded & Edge AI: Model compilation and quantization for NPU deployment (Hailo DFC), edge inference optimization, real-time sensor fusion, embedded C on STM32 (HAL, interrupt-driven UART/I2C)

Computer Vision & ML: PyTorch, Ultralytics YOLO, pose estimation, object detection, OpenCV, DepthAI, model calibration and quantization, ONNX

Robotics & Systems: ROS2, SLAM (slam_toolbox), path planning (A*), sensor integration (LiDAR, stereo depth, ToF), state machine design, microROS

Python Ecosystem: NumPy, pandas, scikit-learn, Matplotlib, FastAPI, Docker

Other Tools: Git, MATLAB, VS Code, Linux, technical writing (IEEE LaTeX)

EXPERIENCE

Robotics Research Assistant

May 2025 – Present

University of Guelph, Dept. of Electrical and Computer Engineering

Guelph, Ontario

- Built a real-time **fall detection pipeline** for an indoor assistive robot, compiling a pose model (YOLO26n-pose) to Hailo Executable Format and deploying it on a Hailo-10H NPU (40 TOPS) for on-device inference with no cloud dependency.
- Diagnosed a model quantization failure that collapsed class confidence output and fixed it via representative calibration data, then implemented software-side NMS and keypoint decoding to support a model head incompatible with native on-chip NMS.
- Integrated a stereo depth camera for synchronized RGB-D input and built a ROS2 publisher feeding live person-tracking data into a collaborator's LiDAR-based path planning system for obstacle avoidance.
- Co-authored an **IEEE-format conference paper** on the system architecture and evaluation, restructured a multi-contributor GitHub codebase with setup documentation across hardware subsystems, and maintained detailed weekly logs documenting project progress and technical findings.

PROJECTS

LLM-Powered Medical Research Paper Summarizer | *Python, FastAPI, spaCy, Docker*

- Built an end-to-end **medical document processing pipeline** using PyMuPDF, pdfplumber, spaCy, and tiktoken to extract and clean scientific text, mirroring the data ingestion and feature engineering work involved in processing clinical records.
- Designed a **map-reduce LLM pipeline** with token-aware chunking, fallback logic, and retries across Claude and GPT APIs, demonstrating experience building robust, production-grade data and modelling pipelines.
- Implemented **automated model evaluation** using ROUGE and BERTScore metrics with structured performance logging to rigorously assess output quality and track accuracy across iterations.

Endangered Species Visualizer | *React, TypeScript, Tailwind CSS, Mapbox API*

- Collected, cleaned, and consolidated 250+ records from dispersed datasets, performing exploratory analysis to identify patterns, quality issues, and key features for filtering and geospatial visualization.
- Built interactive dashboards and reporting tools to communicate complex analytical findings clearly to public and academic audiences, demonstrating experience translating data outputs into usable, interpretable results.

Gameboy Emulator | *C, SDL2*

- Engineered a fully working Game Boy emulator in **C**, replicating CPU, memory, graphics (PPU), audio (APU), and input subsystems with cycle-accurate instruction decoding for full ROM compatibility.
- Architected a modular system design with **SDL2** for real-time graphics rendering, enhancing maintainability and enabling scalability for additional emulator features.